

Reflection Journal

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Learning Reflection

This lab helped me understand that machine learning depends on several tools working together rather than on one library alone. Before working with the Iris dataset, I viewed pandas, NumPy, and matplotlib

mostly as separate Python libraries. Working through the notebook made their roles clearer.

Pandas was

useful for organizing the dataset into a DataFrame and grouping information by species, while NumPy

made numerical calculations such as the mean and standard deviation straightforward.

Matplotlib added

another way of understanding the information because I could see patterns instead of relying only on

printed numbers.

One useful insight came from comparing the numerical results with the scatter plot. The Iris dataset

contains 150 samples, four features, and three classes. The visualization made it easier to notice that the

setosa observations are more clearly separated from the other species, while versicolor and virginica have

more overlap. This showed me why visualization is important in data analysis: a table of values can provide

exact information, but a graph can reveal patterns and relationships much more quickly.

I also became more comfortable with the idea of using different tools for different parts of a machine

learning workflow. The DataFrame made the data easier to inspect and analyze, NumPy handled numerical

operations, and matplotlib turned the results into a visual explanation. The basic statistical analysis also

showed me that averages can summarize a dataset, but they do not tell the entire story. Looking at the

distribution of the observations is important because two groups can have similar summary statistics while

still showing different patterns.

Another important learning point was working with code that already contained a structure and verification steps. The notebook used printed results and checks to confirm that calculations produced the

expected types and outputs. Seeing a message such as "Task 1 completed successfully" helped me

understand the value of testing code instead of assuming that it worked. This also made error checking feel

less intimidating because each result could be examined before moving to the next task.

The GitHub section connected the technical work to a real project workflow. I learned that version control is not only about storing code; it also provides a way to organize project files, track changes, collaborate, and keep a record of work. The idea of committing changes as snapshots was especially useful because it gives a project a history instead of leaving only one final version. Overall, this lab changed how I think about the beginning of a machine learning project. I now see data preparation, numerical analysis, visualization, documentation, and version control as connected parts of the same workflow. One question I would like to explore further is which Iris features are most useful for predicting the species and how accurately a machine learning classifier could separate the three classes. That question connects the exploratory work in this lab to the modeling work that comes next.

EBUKA

Reflective Journal

Completing this lab helped me better understand how the different tools used in machine learning work together. Before working through the exercises, programs such as Pandas, NumPy, Matplotlib, and Jupyter Notebook seemed like separate tools that I would have to learn individually. As I worked with the Iris dataset, I started to understand that each tool has a specific purpose within the same overall process. Jupyter Notebook provides the environment to write and test the code, Pandas organizes the data, NumPy performs calculations, and Matplotlib helps turn the data into something that can be visually interpreted. Seeing these tools interact made the machine learning workflow seem less intimidating because I could understand the role each part played. One of the biggest things I learned was how useful Pandas can be for working with structured data. The DataFrame reminded me of a spreadsheet because the information was organized into rows and columns, but I began to see that Pandas provides much more control over the data. For example, I could select a specific column, separate records based on species, calculate averages, or count values without manually going through each row. At first, commands such as `df['species']` or filtering the DataFrame based on a condition did not feel as natural as selecting cells in Excel. However, after seeing the results, the logic became clearer. Instead of manually finding information, I could tell Python exactly what group or column I wanted to examine. This

helped me understand why Pandas is valuable when datasets become much larger than something

that could be reviewed by hand.

My understanding of NumPy also improved when I used it to calculate the mean and standard deviation of the sepal lengths. I had previously thought of NumPy mostly as another Python library that had to be imported, but this exercise showed me how it connects programming with mathematical analysis. Instead of calculating each individual value, NumPy could work with an entire collection of measurements at once. This made the idea of performing mathematical operations on data feel much more practical. I can see how this becomes important in machine learning because models may need to process thousands or even millions of values. Being able to

perform calculations across an entire set of data is much more efficient than trying to handle each

value separately.

I also became more comfortable with the idea of using an interactive environment such as Jupyter Notebook. Being able to run one section of code and immediately see the result made it easier to understand what each command was doing. If something did not produce the result I expected, I could focus on that individual section instead of trying to troubleshoot an entire program at once. This approach made Python feel more manageable and encouraged me to experiment with the code rather than being afraid of making a mistake.

Overall, the most important thing I learned from this lab was that machine learning is built from several smaller skills that work together. Organizing data, performing calculations, visualizing relationships, and documenting the results are all necessary before moving into more advanced machine learning techniques. I now have a better understanding of why these tools are commonly used together and how they can help turn raw information into something that can be analyzed. As I continue into more advanced labs, I think this foundation will make it easier to understand not only how to run machine learning code, but also why each step in the process matters.

SAM

Reflective Journal: Module 02 Lab Exercise

Entering the world of machine learning development tools initially felt overwhelming, as I was tasked with understanding pandas, NumPy, and Matplotlib simultaneously. My first challenge wasn't necessarily writing the Python syntax from scratch, as much of the code was provided, but rather deciphering what the code was actually doing to the data behind the scenes. Initially, the code blocks looked like dense text, but as I executed the pandas cells, I realized the output was essentially a highly powered version of an Excel spreadsheet. Seeing the raw data transform into structured, readable rows and columns helped bridge the gap between abstract code and practical data manipulation.

Instead of just blindly running the provided code, I wanted to understand how the parameters actually controlled the visual output. When working through the Matplotlib section, I

experimented with the plot properties by successfully changing a graph's color from blue to purple. This hands-on adjustment was a minor tweak, but it yielded a major realization. It helped me understand that these visualization libraries aren't just operating by magic in the background; they are highly customizable tools where simple syntax changes give you direct, granular control over the final presentation of your data.

Beyond the code itself, navigating a new cloud-based development environment presented its own learning curve. I initially struggled with the Google Colab workspace, specifically when it came time to export my completed notebook so that all the visualizations, code, and cell outputs remained intact in a PDF format. By exploring the Colab file menu and learning the proper print-to-PDF workflow, I realized that mastering ML tools requires more than just understanding Python. It also requires knowing how to effectively navigate and utilize the environments where these models are built, documented, and shared. Overall, this lab shifted my perspective from simply executing scripts to actively analyzing how data flows through a machine learning pipeline.

Cersidy

One of the most important things I learned was the value of data visualization. By looking at the Iris dataset through a scatter plot, I could see that the setosa species was more clearly separated from the other species, while versicolor and virginica had more overlap. This showed me that numbers and statistics are useful, but visualizations can sometimes reveal patterns much faster and more clearly.

The lab also improved my understanding of how different steps fit into a machine learning workflow. I learned that summary statistics provide useful information about a dataset, but they do not tell the whole story. It is important to also examine how the data is distributed and how different groups compare with one another.

Another valuable lesson was the importance of testing and verifying code. The notebook included checks and confirmation messages to make sure that calculations produced the expected results. This taught me not to simply assume that my code is working, but to test my results and investigate errors when they occur.

Overall, this lab gave me a better understanding of the basic data analysis process and how Python tools work together in machine learning. It also reinforced the importance of combining numerical analysis, visualization, and testing when working with data. These are skills I can build on as I continue learning machine learning and